MESC-DAST: A Unified Framework for Medical Image Classification and Grading

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Abstract. Medical image classification and grading remain challenging due to low contrast, complex lesion morphology, and pronounced multi-scale structural variations. In addition, ordinal medical grading tasks exhibit structured inter-class relationships that are not well captured by conventional one-hot supervision. To address these issues, we propose a unified framework that combines a Median-enhanced Spatial and Channel Attention (MESC) module with a Distance-Aware Soft Target (DAST) loss. The MESC module integrates average, maximum, and median statistical cues for channel recalibration and employs multi-scale depthwise convolutions to model spatial patterns from local textures to global structures. The DAST loss constructs class-distance-decayed soft targets to explicitly encode ordinal relationships, enabling the model to better distinguish adjacent errors from long-distance misclassifications during optimization. Experiments on multiple public datasets, including HAM10000, Chest X-ray Image, ADNI, Brain Tumor MRI, and Knee Osteoarthritis, demonstrate that the proposed method consistently improves over the ResNet50 baseline and achieves competitive performance across both general medical image classification and ordinal grading tasks. In particular, the proposed framework yields better overall classification results on the standard classification datasets and improves grading consistency on Knee Osteoarthritis. These results indicate that combining enhanced feature modeling with distance-aware supervision provides an effective solution for medical image classification and grading.

Keywords: Medical image classification · Medical image grading · Attention mechanism · Soft target loss

1 INTRODUCTION

Medical image analysis is one of the important application directions of Computer Vision and plays a crucial role in disease screening and graded diagnosis. In recent years, Deep learning methods, especially Convolutional Neural Networks (CNNs), have made significant progress in this field [1]. However, different from natural image classification tasks, medical grading problems usually have more complex structural characteristics: on the one hand, lesion regions often exhibit features such as low contrast, irregular morphology, and significant noise interference, which pose higher requirements for the model's feature extraction ability [1][23][27]; on the other hand, different classes usually have a clear ordinal structure, for example, in the Kellgren–Lawrence (KL) grading, the differences between adjacent grades are often subtle, while those between distant grades are clearly distinguishable [2][3][10].

Most existing methods are modeled based on standard CNN backbones (such as ResNet) [4], and enhance feature representation ability by introducing attention mechanisms or multi-scale structures [5][6][7][8][13][14][15][16]. However, such methods typically rely on simple statistics such as average pooling or max pooling to construct attention weights [5][6], which can easily lead to unstable feature estimation in the presence of noise or abnormal responses. Meanwhile, although multi-scale modeling can enhance the model's perception of different structures to some extent [7][8][13][14], it often lacks effective constraints on the relationships between scales during the feature fusion process, resulting in the model's generalization ability still being limited in complex medical scenarios [8][23].

On the other hand, in the design of loss functions, most methods still use one-hot labels combined with cross-entropy loss for training, treating each category as an independent discrete label [9][10]. This modeling approach ignores the ordinal relationship between categories, preventing the model from distinguishing between "adjacent category errors" and "long-distance errors" during the optimization process, thus limiting its performance in medical grading tasks [3][9][10]. Although existing studies have attempted to introduce certain structural information through methods such as label smoothing or ordinal regression [9][10][11], and further combined uncertainty modeling or soft label supervision in medical tasks to alleviate the limitations imposed by hard labels [22][25][26][27], it remains difficult to strike a good balance among expressiveness, structural modeling ability, and implementation complexity [10][23][25].

To address the above issues, this paper proposes a unified optimization framework from two aspects: feature modeling and loss function design. Specifically, we introduce a Median-enhanced Spatial and Channel Attention (MESC) module into the middle and high layers of the residual network. By combining multiple statistical information and multi-scale spatial modeling, we enhance the robustness and discriminative ability of feature representation. Meanwhile, we design a Distance-Aware Soft Target Loss function, which explicitly encodes the distance relationships between classes, enabling the model to better reflect the structural characteristics of medical grading tasks during the training process.

The main contributions of this paper are as follows:

- A Median Enhanced Spatial Channel Attention Module (MESC) is proposed. This module introduces median pooling in channel attention to enhance robustness against noise, and combines multi-scale depth convolution to effectively model spatial features, thereby obtaining more stable and discriminative feature representations in complex medical images.
- A Distance-Aware Soft Target Loss function is proposed. This method constructs a soft target distribution based on category distance decay, enabling

the model to distinguish different degrees of classification errors, thus better meeting the actual requirements of medical grading tasks.
- The effectiveness of the proposed method has been verified on multiple datasets, achieving competitive performance (SOTA). Experimental results show that this method outperforms existing methods in both accuracy and robustness.

2 Related work

Convolutional Neural Networks (CNNs) have long dominated Computer Vision tasks, particularly in medical image analysis, where they demonstrate strong feature extraction capabilities[1]. Represented by ResNet proposed in Deep Residual Learning for Image Recognition, the introduction of residual connections effectively alleviates the vanishing gradient problem in deep networks, enabling a significant expansion of network depth and thus bringing continuous performance improvements [4]. On this basis, subsequent research generally adopts the approach of introducing lightweight modules into the ResNet backbone structure to enhance representational ability, such as embedding feature enhancement units in the middle and high-level feature stages (e.g., layer3, layer4), to improve the modeling ability of complex patterns while controlling computational overhead [5][6][7][8][15]. However, traditional residual structures still essentially rely on local convolution operations, with limited ability to model long-range dependencies and global context; to alleviate this issue, subsequent research has introduced mechanisms such as non-local operations to explicitly capture long-range dependencies [16]. This limitation is particularly prominent in medical image scenarios, as lesion regions typically exhibit characteristics such as cross-scale distribution, low contrast, and strong noise interference [1][23][27].

To address the limitations of CNNs in global modeling, attention mechanisms have been widely introduced into visual models. Channel attention methods, such as Squeeze-and-Excitation Networks, recalibrate features by modeling inter-channel dependencies [5]; spatial attention methods, such as CBAM, further incorporate spatial dimension information to enhance the response of key regions [6]. These methods have achieved significant performance improvements in natural image tasks and have gradually been transferred to medical grading tasks, such as attention-based KL grading models and multi-scale knee osteoarthritis grading networks that incorporate attention mechanisms [7][8]. In addition, for the automatic grading of knee osteoarthritis, researchers have also proposed frameworks based on Siamese CNN, ensemble learning, and multi-stage processing workflows to improve the recognition accuracy and stability of KL grading [18][19][20][21]. However, existing attention mechanisms typically rely on limited statistical descriptions, such as average pooling or max pooling, to generate attention weights [5][6]; in the presence of noise or abnormal responses, this design is prone to unstable attention estimation. Moreover, their spatial modeling capabilities are often limited to a single scale or shallow convolution structures, making it difficult to fully characterize the multi-scale structural features in complex medical images [6][7][8][13][14].

Multi-scale modeling, as an important direction for enhancing the expressive power of visual models, is particularly crucial in medical image analysis because lesion regions often exhibit significant scale variations [1]. Early methods such as the Inception series achieved feature fusion through parallel multi-scale convolution branches [13], while subsequent work more often adopted depthwise separable convolution to reduce computational complexity while maintaining expressive power, with typical representatives including Xception and MobileNetV2 [14][15]. In recent years, some studies have attempted to combine multi-scale convolution with attention mechanisms to achieve better performance in complex scenarios [7][8][20][22]. For example, in the knee osteoarthritis grading task, the method based on attentive multi-scale CNN has demonstrated that the fusion of multi-scale and attention helps improve the representation ability for different structural scales [8]. However, existing methods often lack effective structural constraints during the multi-scale feature fusion process, making it difficult to adaptively adjust the contributions between features of different scales, and they are still relatively sensitive to noise in high-resolution medical images, which limits their stability and generalization ability in actual clinical scenarios [8][20][23].

In medical grading tasks, especially Kellgren–Lawrence grading, there is usually a natural ordinal structure among classes [2][3][10][18][19][20][21][22], yet this structure is often overlooked in traditional classification frameworks. Mainstream methods typically use one-hot labels combined with cross-entropy loss for training, causing the model to treat each class as an independent discrete label [9][10]. To alleviate this issue, some studies introduce label smoothing to soften the label distribution, thereby reducing the risk of overfitting [11], but it still fails to capture the structural relationships among classes; Focal Loss improves model performance by reweighting hard samples [17], but also does not explicitly model the distances between classes; while methods based on ordinal regression or ordinal regularization improve medical grading performance by explicitly utilizing class order information [3][9][10], but usually introduce additional modeling complexity. Overall, although these methods improve classification performance to some extent, they still fail to effectively model the "distance semantics" between classes and struggle to distinguish between adjacent class errors and long-distance errors.

Apart from the relationship of category order, medical image annotation itself is often accompanied by significant inter-observer variability and uncertainty. Related research has pointed out that inter-observer variability in medical image tasks directly affects the reliability of Model Training, performance evaluation, and uncertainty estimation [24]. In response to this issue, the field of medical imaging has begun to systematically study uncertainty estimation in classification tasks in recent years [23], and has explored the use of soft labels or multi-annotator probability distributions to represent annotation ambiguity and clinical uncertainty in tasks such as segmentation [25][26]. Additionally, some work has improved the robustness and interpretability of medical image evaluation systems in real-world scenarios by explicitly modeling prediction uncertainty [27]. Although these studies provide important insights for uncertainty modeling, there is still a lack of a concise and effective solution for simultaneously and uniformly expressing "category order", "category distance semantics", and "annotation uncertainty" in medical grading problems [9][10][23][25][26][27].

Based on the above analysis, existing methods have deficiencies in both feature modeling and loss design: on the one hand, attention and multi-scale modeling methods lack explicit modeling of noise robustness; on the other hand, classification loss fails to fully utilize the ordered structure between classes and has difficulty naturally expressing the annotation uncertainty in medical grading [3][9][10][23][24][25][26][27]. To address these issues, this paper proposes an attention mechanism that combines robust statistics and multi-scale modeling, as well as a soft label loss function that explicitly encodes class distance information, from the perspectives of feature enhancement module and loss function design, respectively, so as to achieve more stable and clinically cognizant predictions in complex medical grading tasks.

3 Method

3.1 Overall Framework

The proposed medical image classification framework aims to effectively model multi-scale structural features in complex medical scenarios, characterize the ordered relationships and uncertainties between classes, and thereby enhance the discriminative ability of the model in grading and diagnosis tasks. The framework consists of two key components: the Median-enhanced Spatial and Channel Attention (MESC) module and the Distance-Aware Soft Target (DAST) Loss. Specifically, the MESC module constructs robust channel attention by fusing mean, maximum, and median statistical information, and combines multi-scale depth convolution to achieve spatial feature modeling, thereby capturing key medical features from local texture to global structure while suppressing noise interference; the DAST Loss introduces structural information between classes by constructing a soft label distribution based on class distance attenuation, enabling the model to distinguish different degrees of classification errors, allowing for the ambiguity of adjacent classes while strengthening the penalty for misclassification at a long distance, thus better aligning with clinical cognition and annotation uncertainty in medical grading tasks.

3.2 Median-enhanced Spatial and Channel Attention (MESC)

The complete architecture of the proposed MESC module is illustrated in Fig. 1. Given the input feature map:

$$F \in \mathbb{R}^{C\times H\times W} \tag{1}$$

where C, H, and W denote the number of channels, height, and width of the feature map, respectively.

The design of the MESC module directly stems from two key challenges in medical images:

- Noise and artifacts lead to statistical instability (e.g., imaging noise in X-ray, artifacts in MRI)
- Lesion structures exhibit significant multi-scale characteristics

Channel Attention: Robust Statistical Modeling for Medical Noise In medical images, background tissues, imaging noise, and device differences often introduce a large amount of non-structural interference, making traditional statistical descriptions based on mean or maximum values prone to bias. For example, in KL grading, soft tissue noise around the joint area may affect the average response, while local abnormal brightness may dominate the maximum response.

To this end, we introduce median pooling, which is statistically more robust to outliers, thereby enabling more stable characterization of channel responses. Specifically, we calculate three types of statistics:

$$f_{\text{avg}}, \quad f_{\max}, \quad f_{\text{med}} \tag{2}$$

where f_{avg}, $f_{\max}$, and f_{med} denote channel-wise average pooling, max pooling, and median pooling results, respectively.And perform fusion through a shared mapping function:

$$A_c = \sigma\big(\mathcal{M}(f_{\text{avg}})\big) + \sigma\big(\mathcal{M}(f_{\max})\big) + \sigma\big(\mathcal{M}(f_{\text{med}})\big) \tag{3}$$

where $\mathcal{M}(\cdot)$ denotes a shared mapping function (implemented as a lightweight MLP), $\sigma(\cdot)$ is the sigmoid activation function, and A_c represents the channel attention weights.

This design can be understood as complementary modeling of three types of statistical information: the mean provides the overall trend, the maximum captures significant responses, and the median provides a stable estimate robust to noise, thereby obtaining a more reliable attention distribution in complex medical environments.

Spatial Attention: Multi-scale Modeling for Lesion Structure Lesions in medical images typically exhibit significant scale variations, and discriminative features in different tasks often coexist at both local and global levels. This cross-scale feature distribution makes it difficult for convolution operations relying solely on a single receptive field to fully model key discriminative information. To address this issue, we adopt a multi-scale depthwise convolution structure:

$$F_s = \sum_i D_i(F_0) \tag{4}$$

where F_0 denotes the input feature map, and $D_i(\cdot)$ represents depthwise convolution with different kernel sizes (i.e., different receptive fields). F_s denotes the aggregated multi-scale feature.

Based on this, we generate spatial attention maps and reweight the features:

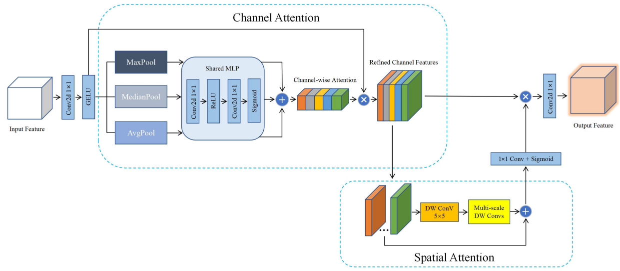


Fig. 1: Illustration of the Median-enhanced Spatial and Channel Attention (MESC) module.

$$F'' = A_s \odot F_s \tag{5}$$

where A_s denotes the spatial attention map, and $\odot$ represents element-wise multiplication. F'' is the refined feature after spatial reweighting.

Thereby achieving self-adaptive enhancement of key lesion regions, enabling the model to more accurately focus on diagnostically significant structural regions.

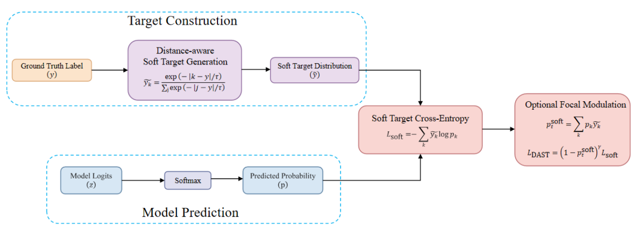


Fig. 2: Illustration of the Distance-Aware Soft Target (DAST) loss.

3.3 Distance-Aware Soft Target Loss

Medical grading tasks typically have a well-defined ordered structure, accompanied by non-negligible annotation uncertainty. Traditional cross-entropy loss treats each category as independent and fails to capture this structure.

To address this, as depicted in the computational workflow in Fig. 2, we construct a soft label distribution based on category distance:

$$\tilde{y}_k = \frac{\exp\left(-\frac{|k-y|}{\tau}\right)}{\sum_j \exp\left(-\frac{|j-y|}{\tau}\right)} \tag{6}$$

where y is the ground-truth label, k and j index class categories, and τ is a temperature parameter controlling the smoothness of the distribution. $\tilde{y}_k$ denotes the soft label assigned to class k.

On this basis, we define the final loss:

$$\mathcal{L} = -\sum_k (1 - p_k)^\gamma \tilde{y}_k \log p_k \tag{7}$$

where p_k denotes the predicted probability for class k, and γ is the focusing parameter that emphasizes hard samples.

This loss explicitly models inter-class distance, enabling the model to tolerate ambiguity between adjacent classes while penalizing large misclassifications, thus improving robustness and generalization in medical grading tasks.

4 Experiment

4.1 Datasets and Evaluation Metrics

Table 1: Comparison between MESC & DAST and classification model baseline.

Method	Bone Disease		Lung Diseases		Brain Diseases			
	Knee Osteoarthritis		Chest X-ray Image		ADNI		Brain Tumor MRI	
	ACC	F1-score	ACC	F1-score	ACC	F1-score	ACC	F1-score
VGG16	66.00	0.6600	92.22	0.9194	98.00	0.9801	90.97	0.8970
VGG19	64.00	0.5800	95.04	0.9504	97.11	0.9712	94.83	0.9470
Xception	-	-	86.02	0.8526	82.66	0.8265	91.79	0.9018
Inception-ResNetV2	63.00	0.6000	86.51	0.8653	89.65	0.8739	90.66	0.9077
InceptionV3	-	-	86.82	0.8479	68.59	0.7017	88.15	0.8760
DenseNet121	64.00	0.5900	84.27	0.8020	89.53	0.8980	91.57	0.9041
DenseNet169	-	-	87.23	0.8652	95.94	0.9594	92.23	0.9223
DenseNet201	-	-	86.37	0.8731	92.34	0.9243	91.74	0.9174
MobileNetV2	67.00	0.6700	86.68	0.8576	68.28	0.6824	91.73	0.9160
ViT	-	-	89.82	0.9000	93.70	0.9210	84.71	0.8500
ResNet101	69.00	0.6500	95.36	0.9538	96.48	0.9726	88.02	0.8620
ResNet50V2	-	-	89.02	0.8890	60.62	0.6034	90.77	0.9081
ResNet50	66.12	0.6659	95.73	0.9562	94.45	0.9248	94.94	0.9482
ResNet layer3+MESC+DAST	**69.99**	**0.6956**	**96.51**	**0.9680**	**98.91**	**0.9816**	**95.62**	**0.9554**

Data sources: Knee Osteoarthritis [28], Chest X-ray Image [29, 30], ADNI [31–34], Brain Tumor MRI [35–39].

Datasets We conduct experiments on five publicly available medical imaging datasets, covering diverse clinical scenarios, including skin lesion classification, chest X-ray disease recognition, Alzheimer's disease classification, brain tumor MRI classification, and knee osteoarthritis (KOA) severity grading. These datasets are selected to evaluate the robustness and generalization ability of the proposed method across different imaging modalities and task types. In particular, the first four datasets correspond to standard multi-class classification problems, while the KOA dataset represents an ordinal grading task with inherent class ordering. The five datasets include HAM10000 (dermoscopic images with seven diagnostic categories), Chest X-ray Image (three classes: COVID-19, pneumonia, and normal), ADNI (Alzheimer's disease classification), Brain Tumor MRI (four classes), and a Knee Osteoarthritis dataset annotated with KL grades 0–4. These datasets exhibit varying characteristics in terms of image modality, class distribution, and task difficulty, providing a comprehensive benchmark for evaluating model performance under heterogeneous medical conditions. To ensure fair comparison and reproducibility, we adopt a unified data partitioning strategy across all datasets. Specifically, for datasets without predefined splits, we apply stratified sampling to construct training, validation, and test sets while preserving class distributions. For datasets with official train/test splits, we follow the original partition and further derive a validation set from the training portion. In all cases, the final splits are approximately aligned to a common ratio (around 70%/10%/20%), ensuring consistent experimental settings across datasets.

Evaluation Metrics We evaluate model performance using two standard metrics: Accuracy and F1-score. Accuracy measures the overall classification correctness, while F1-score provides a more balanced assessment by jointly considering precision and recall, making it more suitable for medical imaging tasks with class imbalance. This combination ensures both global performance evaluation and sensitivity to minority classes across all datasets.

4.2 Comparative Evaluation with Baselines

Experimental Setup To evaluate the effectiveness of the proposed method, we conduct comparative experiments against a ResNet50 baseline, our full model (ResNet50 + layer3 + MESC + DAST), and several representative approaches under a unified experimental protocol. All models use ResNet50 as the backbone with input images resized to 224×224. The proposed modules are inserted after the *layer3* stage to enhance mid-to-high level feature representations. For fair comparison, all models are trained under identical settings, including the optimizer, learning rate schedule, batch size, and number of epochs. Specifically, we use the Adam optimizer with an initial learning rate of 1×10^{-4}, a batch size of 32, and train for up to 60 epochs with a OneCycleLR scheduler. Early stopping with a patience of 15 epochs is applied based on validation performance to prevent overfitting and retain the best model.

Result Analysis As shown in Table 1, the proposed model achieves consistently competitive performance across four datasets covering both standard classification and ordinal grading tasks. Compared with the ResNet50 baseline, our method yields stable improvements on Chest X-ray Image, ADNI, and Brain

Tumor MRI, indicating its effectiveness across different imaging modalities and classification scenarios. This performance trend is consistent with the design motivation of MESC, which enhances feature representation through robust channel recalibration and multi-scale spatial modeling, thereby improving the model's ability to capture discriminative patterns under challenging conditions such as low contrast, structural complexity, and inter-class similarity. On the Knee Osteoarthritis dataset, the proposed method also achieves further improvement. This result is particularly meaningful because KOA grading has an inherent ordinal structure, where adjacent grades are often more ambiguous than distant ones. In this case, the distance-aware supervision introduced by DAST provides a better match to the task characteristics by explicitly modeling inter-class proximity, which helps reduce unreasonable long-distance errors and leads to more consistent grading predictions.

4.3 Ablation experiment

Experimental Setup To further verify the effectiveness of each component in the proposed method, ablation experiments were conducted on five medical imaging datasets, namely Knee Osteoarthritis, HAM10000, Chest X-ray Image, ADNI, and Brain Tumor MRI, under the following four settings:

1. ResNet Baseline;
2. ResNet Baseline + DAST;
3. ResNet layer3 + MESC;
4. ResNet layer3 + MESC + DAST.

Among them, Baseline + DAST is used to verify the role of distance-aware soft label loss in different tasks, layer3 + MESC is used to verify the enhancement effect of the mid-high level feature enhancement module on the representation ability of medical images, while the complete model is used to analyze the complementarity when the structural module and loss function are used in combination. Through this disassembly approach, it is possible to more clearly illustrate the contributions brought by the two core improvements proposed in this paper, namely MESC and DAST, respectively, and whether their combination can further improve the model performance.

Table 2: Ablation experiments of MESC+DAST.

Method	Bone Disease		Skin Disease		Lung Diseases		Brain Diseases			
	Knee Osteoarthritis		HAM10000		Chest X-ray Image		ADNI		Brain Tumor MRI	
	ACC	F1-score	ACC	F1-score	ACC	F1-score	ACC	F1-score	ACC	F1-score
ResNet Baseline	66.12	0.6659	88.32	0.8075	95.73	0.9562	94.45	0.9248	94.94	0.9482
ResNet Baseline + DAST	67.03	0.6691	89.07	0.8288	96.12	0.9630	93.28	0.9283	95.19	0.9506
ResNet layer3 + MESC	68.84	0.6762	89.62	0.8337	96.51	0.9679	98.59	0.9787	95.19	0.9507
ResNet layer3 + MESC + DAST	**69.99**	**0.6956**	**89.27**	**0.8368**	**96.51**	**0.9680**	**98.91**	**0.9816**	**95.62**	**0.9554**

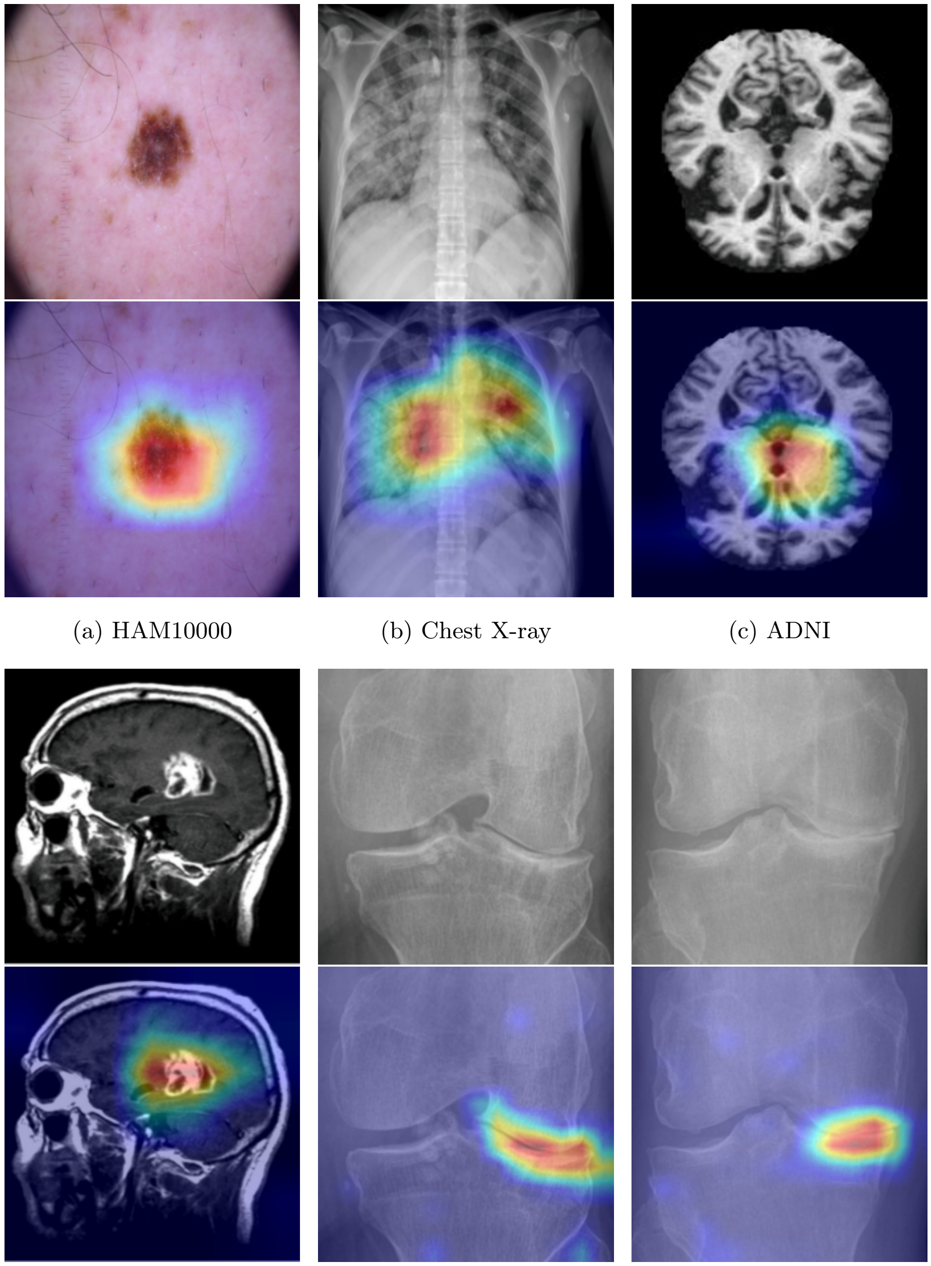
Result Analysis As shown in Table 1I, both MESC and DAST consistently improve performance across datasets, while their effects are complementary. On general classification tasks, MESC mainly enhances feature representation by improving mid-to-high level semantic modeling, enabling the model to better capture discriminative patterns and suppress irrelevant background noise. This leads to more robust performance under challenging conditions such as class imbalance, low contrast, and complex image structures. In contrast, DAST primarily improves the training supervision by introducing soft label distributions, which help the model learn smoother and more flexible decision boundaries, resulting in more stable and balanced predictions across classes.

On the knee osteoarthritis grading task, the effect of DAST becomes more pronounced, as modeling inter-class relationships is crucial for handling ordinal labels with ambiguous boundaries. By explicitly encoding category proximity, DAST allows the model to better tolerate small deviations while reducing severe misclassifications. Meanwhile, MESC contributes by enhancing structural feature representation of joint regions, further supporting accurate grading. As a result, the full model achieves the best or most stable performance across all settings, demonstrating that combining feature enhancement (MESC) and supervision optimization (DAST) leads to consistent and complementary gains across different tasks.

4.4 Grad-CAM Analysis

To further investigate the interpretability of the proposed model, we visualize representative samples from five datasets using Grad-CAM, as shown in Fig. 3. Across different datasets, the model consistently focuses on clinically relevant regions rather than background noise. For instance, in skin lesion images, high-response regions concentrate on lesion areas and boundaries; in chest X-ray images, activations align with abnormal pulmonary regions; in ADNI images, the model attends to brain regions related to disease-associated structural patterns; and in Brain Tumor MRI, responses are localized around tumor-related regions. These observations indicate that the proposed method effectively captures meaningful visual patterns across diverse imaging modalities.

For the knee osteoarthritis grading task, the activation patterns further reflect sensitivity to structural variations. The model primarily attends to key anatomical regions such as joint spaces and adjacent bone structures, and its focus shifts with disease severity, from subtle localized responses in early stages to more concentrated patterns in advanced cases. This suggests that the model not only learns discriminative features but also captures progression-related cues, providing interpretable support for its predictions.



(a) HAM10000 (b) Chest X-ray (c) ADNI

(d) Brain Tumor MRI (e) Knee (Moderate) (f) Knee (Severe)

Fig. 3: Grad-CAM visualization on five representative datasets. For each example, the original image (top) and the corresponding activation map (bottom) are shown.

5 CONCLUSIONS

This paper proposes a unified framework for medical image classification and grading to address the limitations of existing methods in complex structural representation and inter-class relationship modeling. Specifically, we introduce a Median-enhanced Spatial and Channel Attention (MESC) module, which improves the representation of key lesion regions and structural patterns by integrating multiple statistical cues with multi-scale spatial modeling. In addition, we propose a Distance-Aware Soft Target (DAST) loss, which explicitly incorporates inter-class distance information into training, enabling the model to distinguish adjacent errors from long-distance misclassifications in a more reasonable manner. Experimental results on multiple public datasets, including HAM10000, Chest X-ray Image, ADNI, Brain Tumor MRI, and Knee Osteoarthritis, demonstrate that the proposed method consistently outperforms the ResNet50 baseline and achieves competitive performance on both general medical image classification and ordinal grading tasks. Further ablation studies verify the complementarity between MESC and DAST: the former mainly enhances mid- and high-level feature representation, while the latter primarily improves supervision signals and decision boundaries. Their combination leads to more stable and comprehensive performance gains. In particular, on the Knee Osteoarthritis dataset, where the ordinal relationship among classes is explicit, the distance-aware modeling of DAST further improves grading consistency, validating its effectiveness in medical grading scenarios. Overall, the results suggest that, for medical image analysis, simply relying on deeper backbone networks is insufficient to fully address the challenges of complex feature modeling and structured classification. By combining an enhanced attention-based feature modeling mechanism with a distance-aware supervision strategy, the proposed framework provides a more effective solution for improving overall discriminative performance. Future work will focus on lighter module design, stronger cross-dataset generalization, and more rigorous validation in clinically realistic settings to further enhance the practical value of the proposed method.

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